Christopher Marsh

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Data architect and ontologist with experience in healthcare and pharmaceutical research. At NICE I developed a clinical guideline model connecting recommendations to their rationale and supporting evidence, and aligned healthcare concepts to SNOMED CT. At AstraZeneca I developed a pharmaceutical ontology and knowledge graph, working with domain experts and technical teams.

RECENT WORK

2025 – 2026

AstraZeneca

Ontologist

- ✓ Developed an OWL and SHACL ontology for pharmaceutical development, connecting concepts across materials science, formulation and laboratory operations.
- ✓ Created a graph layer in Neo4j for agents to work with synthetic lab data; I grounded the graphs in the ontology so that everything connected together.
- ✓ Set up versioning and release control for the ontology in GitHub, using GitHub Actions to run automated checks before release.
- ✓ Wrote Python transformations to convert relational models into ontology-based semantic data products; this made analytical queries between different teams' data possible for the first time.
- ✓ Worked directly with R&D scientists and ontologists, translating competing requirements and levels of data maturity into an intuitive, usable model.

2020 – 2024

National Institute for Health and Care Excellence (NICE)

Senior Data Architect

- ✓ Developed a graph model of NICE guidelines linking recommendations to committee discussion, technical analysis and source evidence, enabling readers to identify the rationale and its support.
- ✓ Aligned healthcare concepts in the guideline model to SNOMED CT to improve retrieval.
- ✓ Developed NICE's data strategy and enterprise data architecture, combining central governance and standards with distributed data ownership, and designed a semantic data fabric to connect datasets across the organisation and wider healthcare ecosystem.
- ✓ Managed a team of data architects, data engineers and taxonomists; worked closely with information governance.

2020 – 2020

UK Atomic Energy Authority (UKAEA)

Data Strategy consultation

- ✓ Developed a strategy and capability maturity model supporting UKAEA's ambition of FAIR data at scale, advising senior stakeholders on a governance programme for its scientific and engineering 'big data'.
- ✓ Evaluated how semantic models and engineering information standards could support digital-twin objectives and longer-term data interoperability.

2019 – 2019

Royal Pharmaceutical Society (RPS)

Knowledge Modelling consultation

- ✓ Developed an OWL ontology for products, classifications and structured reference content, aligned to healthcare terminologies and vocabularies, and ran workshops for subject experts and developers on taxonomy management.

2015 – 2019

Independent consultant

Data and content architecture

- ✓ Worked with clients including the Royal Opera House and BBC. At the Royal Opera House, modelled productions, performances and participants to support management of its archive.

2012 – 2014

Pearson International

Global Director of Content Architecture

- ✓ Led content architecture strategy and metadata governance across the international organisation, including the development of models for learning content and learning objectives.

CORE EXPERTISE

Semantic architecture: Ontologies, knowledge graphs, OWL, SHACL and Neo4j.

Healthcare data: NHS Data Standards, clinical guideline modelling.

PROFESSIONAL DEVELOPMENT

Level 5 Qualified Coach, EMCC-certified (2023–24)

Privacy by Design, Ryerson University (2018)

EDUCATION

1990 – 1993

BA (class I), University of Oxford, Human Sciences

Course covered human genetics, human health and disease, animal behaviour, human ecology.